Optimizing Magnesium Phosphate Binders with Boric Acid for Additive Construction Applications

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ABSTRACT

The authors are proposing a comprehensive research approach and terminology for providing a framework to utilize an ordinary portland cement (OPC) alternative class of materials for both extraterrestrial habitats and terrestrial 3D construction applications, where sand and clay can replace regolith. Magnesia-based binders (MBBs) have shown a great promise for use in different additive construction (AC) applications due to its superior properties such as high compressive strength, rapid setting time, and excellent durability, in addition to the possibility of harvesting these binders from lunar and Martian regolith. Our primary objective in this research is to prove the concept of utilizing magnesia phosphate cement (MPC) composites with different additives, including boric acid, graphene nanoplatelets (GnP), acetic acid, and potentially Martian and lunar regolith simulants as a potential construction material for the planetary habitation exploration missions. Before adjusting the mix design properties required to meet specific 3D construction printing needs, several aspects must be studied in depth to provide a better understanding of the MPC paste and composite behavior. This work aims to give more insight into the relative physical, mechanical, thermal, and chemical performance of the MPC pastes with varying additions of boric acid. The experimental results show that boric acid can change the phase compositions of the MPC paste and affect its performance. It can improve the initial setting time of the paste and alter the structure of its binding products. Finally, our proposed research creatively utilizes artificial neural networks (ANN) to investigate and subsequently optimize both the MPC pastes and composite formulation for real-time adjusting of 3D printing processes based on the desired material performance criteria.

Key Words: Additive Construction, Extraterrestrial, Planetary, Artificial Neural Networks (ANN), Boric Acid, Magnesia Phosphate Cements (MPC), Ordinary Portland Cement Alternatives, In-Situ Utilization.

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INTRODUCTION

The inclusion of Additive Manufacturing (AM) technologies in the construction field has the potential to solve many structural-related problems and will reduce construction cost and time. The three primary challenges that need to be solved to incorporate AM in construction applications are: (1) development of the 3D construction printing (3DCP) process for non-homogeneous materials, (2) improvement of the printing process and (3) the construction durability. Planetary autonomous construction is facing some technology maturation issues, such as the scarcity of readily-available printable construction materials, accompanied by material science challenges particular to the planetary surface environment, production process of the construction material, 3D printing process-related material requirements, construction-related material requirements, and handling of non-homogeneous and time-dependent material properties. Immense work is required to improve the AC processes and to utilize materials that have in-situ development potential.

Utilizing in-situ resources for the construction of habitat structures on the Moon and Mars will increase the efficiency of long-duration space missions and will reduce the number of payloads launched from Earth necessary, thus reducing cost. Hence, there is a critical need to understand the transformation of indigenous planetary substances, such as lunar and Martian regolith, and any respective atmospheric or volatile constituents into revolutionary resilient construction materials (Edmunson et al. 2018, NASA 2015). Also, a need exists to better understand the available in-situ resources as construction materials at specific settlement locations, which will result in optimal cementitious mixes. Along with formulating several optimal cementitious composite mixes customized for specific landing sites, developing these materials will require intensive multiscale/multifunctional characterization and fundamental modeling, and validation of technology transformation (Cesaretti et al. 2014).

Magnesium phosphate cement (MPC) is formed by the reaction between magnesium oxide and a soluble phosphate compound, such as ammonium phosphate monobasic ($NH_4H_2PO_4$) (Khoshnevis 2005). Since both the Moon and Mar's surfaces are rich with magnesium-bearing minerals (Werkheiser et al. 2015), utilizing in-situ materials indicates that MPCs could be a candidate construction binder.

MPCs exhibit many superior properties such as high early compressive strength, rapid setting time, excellent volume stability, exceptional durability, the ability to harden at low temperature, a relatively low drying shrinkage, fire-proof behavior, and superior resistance to abrasion (Li et al. 2014, Yang et al. 2013). Although MPCs exhibit many superior properties compared to OPC, the extremely prompt setting time of a few seconds generally excludes this great in-situ construction material candidate from us in AC technology.

For this research, the authors have identified several additives to reduce the MBBs initial setting time without negatively impacting its mechanical and physical properties. These additives and composite mixes need to be optimized to fit with technology requirements such as those identified by the Additive Construction with Mobile Emplacement (ACME) project (Edmunson et al. 2018, Fiske et al. 2018). The deep-space habitation systems need to balance between the scientific knowledge and need, the architectural demands, and human factors. However, studying this in more depth identifies even more issues and challenges that our research community are not familiar with for our deep-space infrastructure construction mission.

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MAGNESIUM PHOSPHATE CEMENTS

Magnesium phosphate cement (MPC) is a bonding material that belongs to the chemically bonded ceramics (CBCs) group (Formosa et al. 2012). The chemical bonding in MPC is formed by an acid-base reaction between dead burned magnesia and phosphate (Li et al. 2014, Qiao et al. 2010). Traditionally, MPC is used as a repair material for applications such as rehabilitation of civil structures, patching concrete pavement, and fixing damaged runways. Also, MPCs have been used in several environmental and medical applications for wastewater treatment (Masuda et al. 2013) and biomedical implants. Lately, more interest in MPC arises in this acid-base ceramic as a potential 3D printing material for both biomedical implants and complicated structures (Klammert et al. 2010, Inzana et al. 2014).

MPCs are basically formed by mixing dead burned magnesia with phosphate, via diverse sources of phosphate compounds such as phosphoric acid, and ammonium dihydrogen phosphate. In the past, ammonium dihydrogen phosphate (ADP) is mainly used to produce MPC (Li et al. 2014, Qiao et al. 2010, Soudée and Péra 2000, Li et al. 2015). The main MPC reaction product is a crystalline structure known as magnesium ammonium phosphate hexahydrate [$MgNH_4PO_4.6H_2O$] or more commonly known as the mineral struvite (Chau et al. 2011). An unpleasant odor released from the reaction, due to the ammonium phosphate formation, is the main driving reason to replace ADP with potassium dihydrogen phosphate (KDP). A new MPC class was investigated to overcome the odor issue accompanying the struvite formation reaction. Later, it was found that the new MPC binding products contained a better phosphate salt for MPCs. The natural-binding product is a crystalline structure known as magnesium potassium phosphate hexahydrate [$MgKPO_4.6H_2O$] or K-type struvite. This compound has the same structure as struvite, with the replacement of an ammonia ion by a potassium ion (Mathew 1979). The hydration mechanism of MPC to produce the K-struvite compound can be described by the following equation:

$$MgO + KH_2PO_4 + 5H_2O \longrightarrow MgKPO_4.6H_2O \ldots \quad (1)$$

Over the past few years, several researchers studied K-struvite formation and proposed several mechanisms. Qiao et al. (2010) proposed a K-struvite crystal formation mechanism where it dependent on the precipitation/dissolution reaction conditions. At first, magnesia dissolves in water leading to a continuous pH increase of the solution. Then, a crystalline structure known as phosphorrösslerite ($MgHPO_4{\cdot}7H_2O$) begins to form and dissolves in the solution. Finally, another crystalline structure ($Mg_2KH(PO_4)_2{\cdot}15H_2O$) starts to appear and dissolves in the solution to produce the K-struvite crystals. Ding et al. (2012) proposed a mechanism based on the hydrated MPC paste involving two significant compounds: K-struvite and unreacted magnesia. The main contribution from their research is that the K-struvite crystalline form is not necessarily the only possible configuration; it can also exist in amorphous structures. In this case, the unreacted magnesia works as fine aggregates in the hardened matrix leading to better mechanical properties.

Qiao et al. (2012) explained the exothermic reaction that produces the K-struvite. When the magnesia and the monopotassium phosphate dry mix is mixed with water, the phosphates rapidly dissolve in an endothermic reaction. This is followed by two exothermic reactions, the first one is due to the metal-dissolution of magnesia in the acid solution and the second is due to the formation of the K-struvite. In 2017, Le Rouzic et al. (2017) proposed that the dissolution of

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magnesia in an acid phosphate solution takes place, leading to an increasing pH solution, which permits the K-struvite precipitation.

From the mechanisms proposed earlier, fresh and hardened properties of MPCs can be affected by the reactivity of the magnesia and altered by magnesia to phosphate molar ratio (M/P), and water to solid ratio (W/S) in the mix proportions. For example, Xu et al. (2015) studied the effect of M/P molar ratios on the behavior and the microstructure of formed MPC. Their research concluded that high M/P ratios can reduce the mechanical strength and the setting reaction. Conversely, using low M/P ratios can produce a denser microstructure and can lead better packed crystals.

Li et al. (2014) investigated the influence of different M/P molar ratios ranging from three to six on the compressive strength of the MPC system. They reported that the M/P ratio of four is optimal for the compressive strength. Qiao et al. (2010) studied the effects of different M/P ratios ranging from six to twelve on the setting behavior and the compressive strength of MPC pastes. The results indicated that setting time decreases by increasing of M/P ratio. Moreover, as the M/P ratio increases, the compressive strength increases until it reaches an optimum at a ratio of eight, then it decreases with an increasing M/P ratio beyond the maximum.

Other researchers focused on the relation between the M/P ratio and the microstructure of the MPC paste. Ma et al. (2014) studied MPC microstructures by preparing several samples with M/P ratios from four to twelve. The results showed that the lowest porosity, lowest intrinsic permeability, and the highest compressive strength is when the M/P is 6. Similarly, Chau et al. (2011) studied the relation between the M/P ratio and the microstructure of the MPC paste, starting from the phase identification to the morphology examination. According to their study, the final form of K-struvite could be altered based on the growth condition process, which is highly dependent on the M/P ratio. They reported that the mechanical strength of the MPC paste is entirely reliant on the crystallinity of the K-struvite. The crystallinity of the K-struvite is highly dependent on the M/P ratio. The study proved low crystal density at low M/P ratios and better density at high M/P ratios. The best crystalline structure and the best mechanical strength are achieved at an M/P ratio of ten.

The amount of water used to form the MPC pastes is one of the most critical factors that can tailor the binding properties. Li and Chen (2013) studied the influence of the W/S ratio for a series of MPC specimens. They reported that the setting time increased with increasing W/S ratio. Yang and Wu (1999) concluded that mixing with a high W/S ratio could reduce the compressive strength of MPC paste. Yang et al. (2000) reported that the fluidity sensitivity of MPC mortar increases rapidly at a high W/S ratio. Furthermore, they found that the lower the W/S ratio led to higher compressive strength. Fu et al. (2016) investigated the effects of water content on several MPC properties, including temperature evolution, viscosity, and mechanical strength. The results proved that the higher water content, the lower hydration temperature, the higher viscosity values, and the lower mechanical strength.

On the other hand, MPC binders have extremely rapid setting time. To increase the utilization of this cement alternative in infrastructure construction applications, there is a pressing need to control and retard the reaction to obtain the required workability for a reasonable deposition time. Without the presence of a retarder in the MPC mixes, the setting time is in the range of 30 seconds to 15 minutes. The most popular retarder suggested for MPC mixes is borax ($Na_2B_4O_7 \cdot 10H_2O$). Other retardant compounds include boric acid (H_3BO_3) and sodium triphosphate ($Na_5P_3O_{10}$) (STP).

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The influence of three different retarders on the setting behavior of the MPC pastes, including STP, boric acid, and borax are evaluated by Hall et al. (2001). The results indicated that boric acid and borax could prolong setting time, while STP has minimal influence. Unfortunately, Young's modulus and flexural strength decreased using all of these retarders. In agreement, Formosa et al. (2012) research reported that boric acid increased the setting time and improved the workability. Lahalle et al. (2016) studied the effect of boric acid on the dissolution process of MPC compounds; it showed that the boric acid did not affect the dissolution process of the primary compounds, MgO and KH_2PO_4, but it slowed down the precipitation process of the K-struvite.

A series of MPC specimens with different borax contents were studied by Yang and Qian (2010), to investigate the influence of borax on the hydration and the rheological properties. Hydration temperature, pH value, fluidity, setting time, and compressive strength measurements are conducted using X-ray diffraction (XRD) and scanning electron microscope (SEM) techniques. They concluded that borax could decrease the hydration rate-or increase the setting time- by forming a thin film onto MgO surfaces which hinders the dissolution process of these magnesia particles. This retardation mechanism can decrease the water consumption by the primary reactants MgO and MKP; as a result, the fluidity of the system is increased.

The effect of hybrid retarders such as polycarboxylate superplasticizer (PCE) on the rheological properties, flow, and setting time of MPC specimens are studied by Tan et al. (2018). Adding PCE to the MPC mixture could delay the chemical reaction by obstructing the dissolution of MgO and MKP. Whereas, the presence of PCE and borax together in the mix did not improve the workability of the system; the borax presence prevented the absorption of PCE by MgO particles. Nevertheless, the improvement in the setting time (up to 7 minutes), due to the borax addition is sufficient for traditional rapid construction applications (Li et al. 2014). For AC applications it is still not enough, considering an average of an open time operation of 45 minutes.

From the literature presented above, MPC is still facing some unresolved challenges for broader utilization in construction applications. Due to its other superior properties, many researchers resumed their interest and investigated other compatible additives to overcome rapid setting and workability challenges and to improve the overall performance of MPCs in general. Fly ash (FA), silica fume (SF), metakaolin, slag, acetic acid (AA), and graphene oxide (GO) (Li et al. 2015, Zheng et al. 2016, Xu et al. 2017, Li et al. 2017, Lu et al. 2016) are the main additives highlighted in the literature surveyed.

NEEDED RESEARCH

According to the literature provided, several factors can play a significant role in tailoring the mechanical and rheological properties of MPC. These critical factors include the M/P molar ratio, W/S molar ratio, and physical and chemical characteristics of the set retarder. Many contradictions and deficiencies have been found in the literature, understanding the MPC mechanical behavior from varying the mixing and additives proportions and ending with the influence of various retarders. At first, there is an apparent ambiguity in the literature regarding the optimal M/P molar ratio. As some studies have confirmed that the optimal value for M/P ratio is ranging between 4 and 6, others have confirmed that it is more than 10. Also, a few studies examined the relationship between the M/P ratio and the workability.

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MPC mixes are very sensitive to water content, which controls the rheological properties, compared to OPC. In some cases, adjusting water content in the mix to a small extent can make a big difference in initial setting time and flowability, this is shown in the results section. This sensitivity could become a barrier facing the MPC utilization for various infrastructure construction applications, especially for AC. Due to this sensitivity, previous studies could not capture precisely the effect of the water content on the optimal MPC mix ratios. Basing mix calculations on weight ratios rather than molar ratios and treating MPC as OPC paste mixes caused a misinterpretation of the real impact of water reaction with the primary constituents (MgO and MKP) on the fundamental understanding of the MPC behavior.

The rapid setting time of MPC is found to be one of the most significant challenges that is facing its utilization in various applications. However, there are only a few studies focused on the impact of several set retarders on the performance of MPC. Within the thorough literature review search conducted, there is only one study that has discussed the effect of borax in depth. However, there is no published research on the effect of boric acid on the chemical reaction of MPC and its influence on hardened and fresh properties.

METHODOLGY

Utilizing MPC for AC applications requires specific rheological and hardened properties. The most critical properties that can affect the performance of the AC processes are (1) setting time, (2) workability, and (3) green strength. These critical features are correlated with the mix design of the MPC pastes to satisfy the required fresh and hardened properties for 3DCP. Consequently, the experimental program shown in Figure 1 has been proposed to study the effect of boric acid on the performance and microstructure of MPC. Finally, ANN models are utilized to investigate and subsequently optimize both the MPC pastes and the composite mix formulations for desired real-time 3D printing performance requirements.

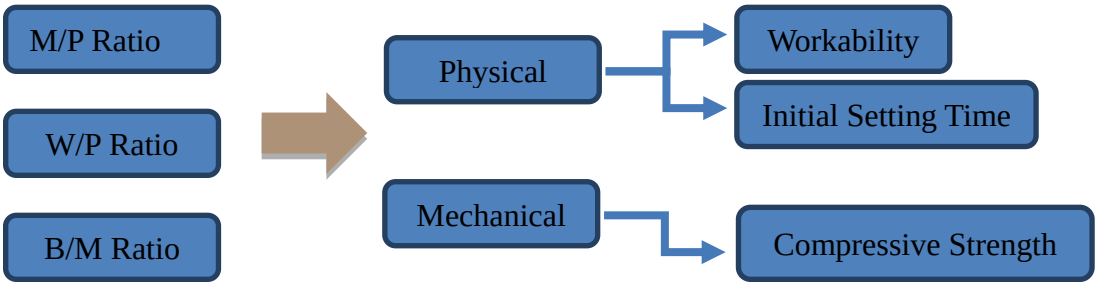

Figure 1: Experimental methodology chart

MATERIALS

The MPC specimens are prepared by mixing dead-burned magnesia (MgO), monopotassium phosphate (KDP), boric acid (H_3BO_3), and distilled water. The used MgO is purchased from Martin Marietta Magnesia Specialties, LLC. Baltimore, Maryland, USA. Table 1 displays the chemical composition of the MgO powder. The KDP is supplied by Premier Magnesia, LLC. Waynesville, North Carolina, USA, and the boric acid is purchased from Duda Energy, LLC, Decatur, AL, USA, with a purity of 99.9%. Table 2 summarizes the physical and chemical properties of the used boric acid powder.

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Table 1: Chemical composition of the dead-burned MgO powder [Martin Marietta Magnesia Specialties]

Magnesium Oxide (MgO)%	Silicon Oxide (SiO_2) %	Calcium Oxide (CaO)%	Iron Oxide (Fe_2O_3) %	Aluminum Oxide (Al_2O_3) %
98.00	0.70	0.95	0.15	0.20

Table 2: Physical and chemical properties of boric acid powder [Duda Energy, LLC]

Physical/chemical property	Description/Measured value
Appearance	Crystalline, White Solid
pH	5.1 at 1.8 g/l at 25 °C (77 °F)
Melting point	160 °C
Initial boiling point and boiling range	1860°C
Relative density	1.440 g/cm3
Solubility	In Water: 4.7% @ 20°C; 27.5% @ 100°C
Decomposition temperature	169±1 to HBO_2 & -1 1/2 H_20 at 300°C

SPECIMEN PREPARATION

As previously mentioned, the behavior of the MPC paste specimens can be driven by three main factors including magnesia to phosphate molar ratio (M/P), water to solid mass ratio (W/S) and boric acid to magnesia mass ratio (B/M). In this paper, the authors chose to represent the MPC pastes ingredients in molar ratios rather than the mass ratios terminology to capture the impact of the different MPC mix proportions (M/P, W/P, and B/M ratios) on the acid-base reaction, in addition to the influence of these factors on physical, and mechanical properties. The MPC specimens were named by character numbers. The first number indicates the M/P ratio; the second two numbers refer to the W/P ratio, and the third two numbers indicate the B/M ratio. For example, B32205 sample indicates a specimen with M/P of 3, W/P of 2.2 and B/M of 5%.

A total of 39 MPC specimens were designed and prepared to study the behavior of the MPC paste under different conditions and divided into eleven different groups. The first seven groups including (B2-20, B3-18, B3-22, B3-30, B6-30, B6-36, and B6-45) were designed and prepared to study the influence of the boric acid on the MPC pastes. For each group, M/P and W/P molar ratios were kept constant, while the B/M ratio was changed for each specimen. The last 4 groups, including W-4, W-8, W-10, and W-12, were prepared to complete the first seven groups to evaluate the influence of M/P and W/P molar ratio on the optimized pastes. For each one of these four groups, M/P and B/M ratios were kept constant, while the W/P molar ratio was changed for each specimen.

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MIXING PROCEDURE

The MPC pastes were prepared using the following procedures: (1) the dry powders were weighed and mixed for 2 minutes, (2) the boric acid was dissolved in the distilled water using a Fisher Scientific Thermix Stirrer Model 120 s with a speed of 300 RPM for 15 minutes. Finally, (3) the water was added to the dry mixture and mixed for two more minutes. After mixing the paste, the MPC fresh mixes were placed into standard 2" cube brass molds following the ASTM standards (C109) for compressive strength testing. After 24 hours, all the MPC specimens were de-molded and cured at room temperature for seven days.

CHARACTERIZATION SCHEME

Physical properties

The Vicat apparatus with a 1 mm needle was used to measure the setting time of the MPC paste samples according to the ASTM C403 standard. Due to the rapid plasticity loss, the initial setting time recording interval was every 30 seconds.

Mechanical properties

The compressive strength for the MPC hardened cubes was measured according to ASTM standard C109, specimens in this research were tested after 7-days of air curing with a loading rate of 0.001 in/sec using an 810 Material Testing System (MTS).

RESULTS AND DISCUSSION

Physical properties of the MPC paste

Influence of boric acid

Figure 2 illustrates the effect of the boric acid content on the initial setting time for the MPC pastes for different W/P and M/P molar ratios. At constant M/P and W/P ratios, as the B/M ratio increases, the initial setting time increases until it reaches the maximum of 5% boric acid content. Increasing the B/M ratio beyond 5% will reduce the setting time. A similar trend obtained for all M/P molar ratios. For example, for B3-22 group (M/P and W/P molar ratios of 3 and 2.2 respectively), adding 5% of boric acid can increase the initial setting time from 3.5 to 15 minutes. Meanwhile, adding 10% to the same mix can increase the initial setting time to 9 minutes only. One possible explanation for this retardation mechanism is the ability of the boric acid to form a temporary layer of lunebergite (Mg_3B_2 $(PO_4)_2(OH)*6H_2O$) which coats the surface of the Mg^{+2} ions and hinder the dissolution of the magnesium cations. This temporary layer can be destroyed when the amount of the magnesium cations is increasing in the solution leading to continuous dissolution of Mg^{+2} particles again and forming the K-struvite. This temporary layer will be dissolved again in the solution and cannot be found in the reaction products of the system (Hall et al. 1998, Formosa et al. 2012). Another possible reason for this retardation is that the lunebergite layer is hindering the precipitation process of the final products (K-struvite) rather than the reactant particles (Mg^{+2}) (Lahalle et al. 2016).

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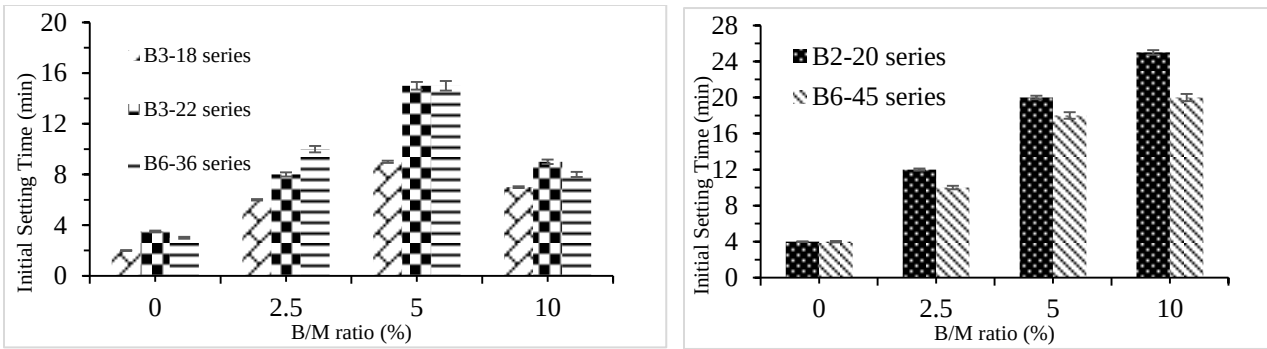

Figure 2: Influence of B/M ratio on the initial setting time of the MPC paste for different W/P and M/P molar ratios

On the other hand, there is an adverse effect on the initial setting time when the B/M ratio exceeds 5%. Increasing the boric acid content increases workability in the mixes. Although the fluidity rate is not measured, during the experiments, the higher the B/M ratio, the more rapidly fluidity loss is observed. This note can explain why the retardation effect of the B/M ratio of 5% is better than the B/M ratio of 10%. To give more insight into the boric acid effect and the fluidity loss of the MPC paste, the additional two groups of MPC pastes were prepared (B2-2 and B6-45). These two groups were prepared with very high-water content to ensure a very high paste fluidity that would delay the setting of the sample. Figure 2b illustrates the effect of the boric acid content on the initial setting time of overflow MPC pastes; the initial setting time increases when the B/M ratio increases. For instance, using 2.5 %, 5%, and 10% boric acid in the case of the B6-45 series can increase the initial setting time by 150%, 350%, and 400%, respectively. One explanation for the fluidity loss could be the negative effect of the boric acid on the dispersion of the Mg^{+2} inside the solution. Another could be the ability of the boric acid to dissolve in the aqueous solution and form an amorphous boron-magnesium compound, in which the water is a significant factor in this formation.

Mechanical Properties Compressive Strength

Influence of magnesia content (M/P ratio)

For each W/P molar ratio, an optimal M/P molar ratio exists. For example, for a W/P molar ratio of three, the highest compressive strength obtained at an M/P molar ratio of six. Whereas, for a W/P molar ratio of 3.8, the optimal M/P molar ratio is ranging between six and eight. Recognizing that the varying of the M/P molar ratio affects the optimal amount of water required for the hardening process. For example, the compressive strength of a W/P molar ratio of 3 and an M/P molar ratio of three is ~1/3 less than when the optimum M/P molar ratio is used. This compressive strength reduction is due to the high-water content, which results in larger voids between the binding particles, and consequently a lower structural density K-struvite. Once the pastes harden, the excessive amount of water will be squeezed out and evaporated, leading to micropore and microvoid formation. For an M/P ratio six, a W/P ratio of three is adequate for the mixture; the additional amount of the magnesia is consuming the excess water and developing more K-struvite which improves the compressive strength. Whereas, for an M/P ratio eight, a W/P ratio of three is insufficient for the magnesia particle dissolution mechanism. This deficiency will increase the number of unreacted magnesia particles and decrease the amount of K-struvite formation, reducing the compressive strength of the MPC specimen. On the other

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hand, reducing water can reduce the flowability of the system leading to poor compaction and weak compressive strength.

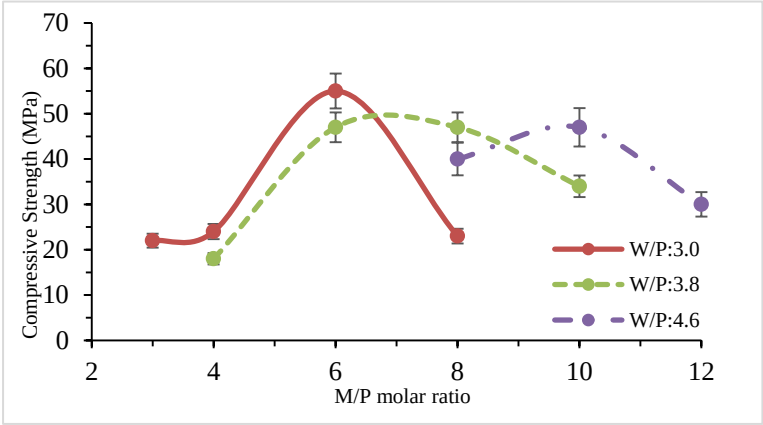

Figure 3: Influence of M/P molar ratio on the compressive strength of MPC paste for different W/P molar ratios

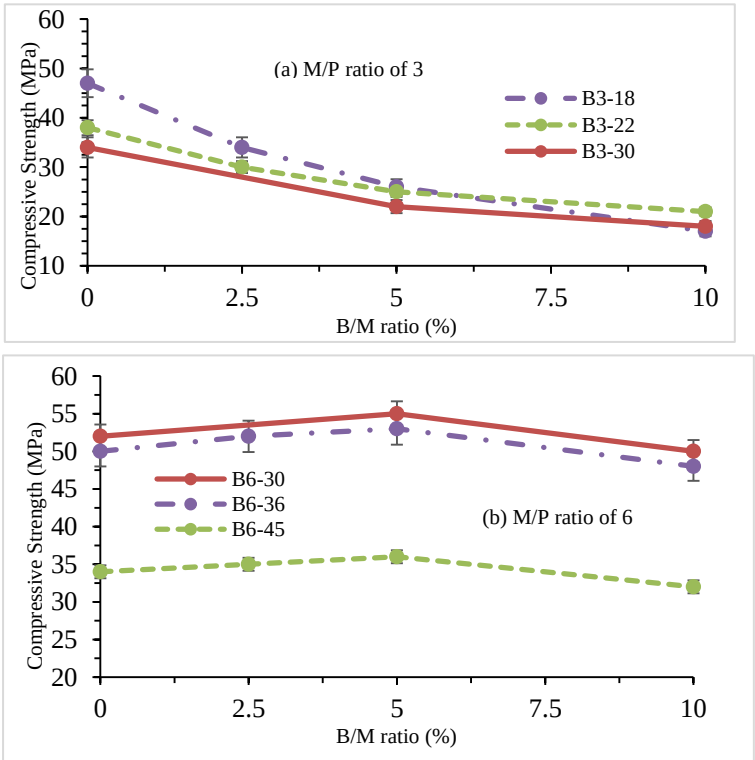

Figure 4: Influence of B/M ratio on the compressive strength of MPC for different W/P molar ratio

In conclusion, the optimal M/P molar ratio depends on the amount of the water used, and this can explain the apparent contradiction found in the literature. Published work primarily expresses the mix of constituents in weight ratios rather than molar ratios, especially for water

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and the KDP. This expression may cause a misinterpretation of the real impact of magnesia/water content on the fundamental understanding of MPC behavior.

Influence of Boric acid

Boric acid content was found to be a critical factor in the MPC pastes compressive strength. Figure 4 demonstrates the influence of the boric acid content on the compressive strength for different M/P molar ratios. The results indicate two different behaviors of the MPC specimens containing boric acid. In the case of low M/P ratios (figure 4a), compressive strength decreases when the B/M ratio increases. For example, B3-18 series (M/P of 3 and W/P of 1.8), adding 2.5% 5% and 10% of boric acid, reduces the compressive strength by 27%, 47%, and 64%, respectively. Adding more water to the MPC mixture, the boric acid effect on the compressive strength was lower. For the B3-22 series, adding 2.5%, 5% and 10% of boric acid reduces compressive strength by 21%, 34%, and 45%, respectively. Another MPC series (B3-30) was prepared to investigate whether the presence of the boric acid had the same influence in the case of high-water content. The results recorded aligned with our initial observation. For the B3-30 series (M/P of 3 and W/P of 3), adding 5% and 10% of boric acid can reduce the compressive strength by 35% and 47%, respectively.

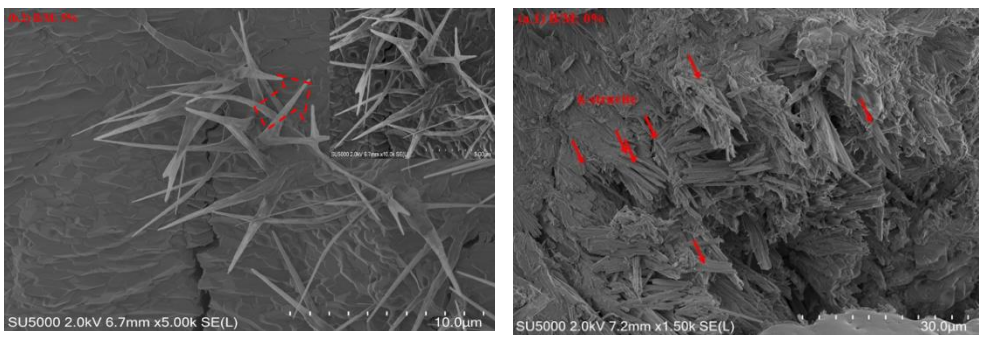

Figure 5: SEM images of MPC paste with low M/P molar ratio; a) B/M ratio of 0%, and b) B/M ratio of 5%

MPC specimens with high M/P ratios (Figure 4b), results show that the boric acid effect on compressive strength is insignificant. Increasing B/M ratio, compressive strength slightly increases, then slightly decreases when the B/M ratio more than 5%. For the B6-36 series (M/P of 3 and W/P of 3.6), adding 5% of boric acid can improve the compressive strength by 6%. Meanwhile, adding 10% of boric acid to the same mixture can reduce the compressive strength by 4%. Furthermore, a similar result was obtained when boric acid was used with high W/P molar ratios.

SEM (Microstructure Imaging)

Microstructure imaging of MPC pastes of low and high M/P ratios are presented in Figure 5. M/P and W/P ratios were fixed at three (right) and 2.2 (left), while the B/M ratio is 0% and 5% respectively. The SEM images of the MPC paste with no boric acid show a dense, mostly crystalline microstructure. A bladed prism of the k-struvite crystalline structure is found dominating. For a 5% boric acid content in the mixture (Figure 5b), two types of microstructures were identified with very different morphologies observed.

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ANN NUMERICAL SIMULATION

For this research, three databases, of 138 total data sets, have been used to develop the ANN models. For each database, two ANN models have been optimized. In the first model the input is the MPC paste or mortar ingredients, and the output is the predicted performance of interest. In the second model the input is the performance of demand (compressive strength, flowability, and initial setting time (not sure I understand this), and the output is the mix ingredients. For this study, a three-layered, feed-forward error backpropagation ANN model was developed utilizing a TR-SEQ1 model. However, due to this publication's space limitation, we will only present the first ANN model results. More work from the same research group will be submitted for publication in the near future.

The ANN1-1 model was developed to predict the fresh and the hardened properties of the MPC pastes including flow, initial setting time, and compressive strength. The input and the outputs of this model are summarized in Table 3. All the data sets are normalized and scaled in a range of 0-1 due to the use of the sigmoidal function. Moreover, a maximum or a minimum of each variable has been included in the training data. The ANN model procedures for training and testing data sets were run five times by changing the number of initial nodes from one to five. Then, the best ANN model is selected based on the Average Square Error (ASE) value of testing data sets. The optimum number of Hidden Nodes (HN) is found to be five, and it is calibrated after 20000 iterations. As a result, the selected ANN model consists of five input nodes in the input layer, five nodes in the hidden layer, and three output nodes in the output layer. The corresponding statistical measurements for this model are ASE_{ts} of 0.0129, R^2_{ts} of 0.84, and $MARE_{ts}$ (Mean Absolute Relative Error) of 31.88 for testing data sets; and ASE_{tr} of 0.0025, R^2tr of 0.93, and $MARE_{tr}$ of 15.85 for training data sets. Then, the model was validated and the statistical measurements for the validation data sets were calculated to be ASE_{val} of 0.013, R^2_{val} of 0.77, and $MARE_{val}$ of 23.6. Table 4 shows the statistical measurements of the ANN1-1 model.

Table 3: Input/outputs of ANN1-1, ANN2-1, and ANN3-1 models.

Input	Outputs
M/P molar ratio	Flow (%)
W/P molar ratio	Initial setting time (min)
B/M ratio (%)	Compressive strength (MPa)

Table 4: Statistical measurements of ANN1-1 model, ANN structure 5-5-3 with 20000 iterations

Model	MARE training	MARE Testing/ validation	R^2 Training	R^2 Testing/ validation	ASE Training	ASE Testing/ validation	ASE Com.
Training-testing	15.85	31.88	0.93	0.84	0.0025	0.0129	0.0154
Training-validation	15.85	23.6	0.93	0.77	0.0025	0.0130	0.0155
Training all	17.28	NA	0.89	NA	0.0042	NA	NA

The comparison between the actual and the predicted data sets of ANN1-1 model for flow, initial setting time, and compressive strength respectively is shown in Figure 6. The graphical accuracy shows that the ANN1-1 model is predicting the MPC properties of interest with a high confidence. For the initial setting time prediction, the agglomeration of points in a range of 0 to

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15 minutes agrees with the nature of the provided data sets. That is, MPC pastes are known for their rapid setting time. Furthermore, the graphical accuracy figures show a good agreement between the actual and the predicted values of the validation data sets. On the other hand, the "training all" technique is applied to improve the accuracy of the selected ANN model. In this technique, all the available data sets are used to train the model while no testing or validation data is presented (Figure 7). The ANN results for the "training all" techniques improved the efficiency of the selected model and enhanced the overall performance. The corresponding statistical measurements for the training all network are $ASE_{(trall)}$ of 0.0042, $R^2_{(trall)}$ of 0.89, and $MARE_{(trall)}$ of 17.28.

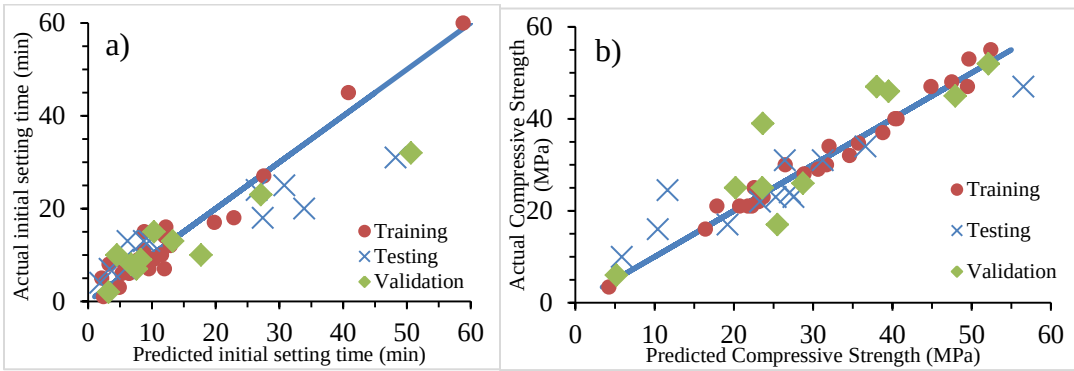

Figure 6: ANN1-1$_{(tr\text{-}ts\text{-}val)}$ model predictions accuracy for a) initial setting time (min), and b) compressive strength (MPa).

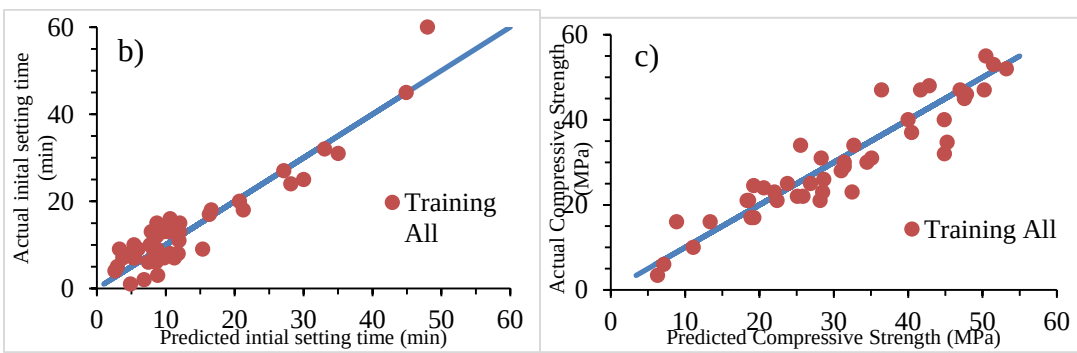

Figure 7: ANN1-1$_{(training\text{-}all)}$ model predictions accuracy for a) flow (%), b) initial setting time (min), and c) compressive strength (MPa).

CONCLUSIONS

The results show that by manipulating the M/P, W/P, and B/M molecular ratios, one can tailor the MPC fresh and hardened properties for 3D printing. Using a molar ratio representation rather than a weight ratio can provide a better understanding of the MPC's sensitive behavior under different formation conditions. The presence of boric acid can affect the physical properties of MPC pastes, decreasing the workability of the fresh mix, and increasing the fluidity loss with time. Based on our results, using up to 5% B/M ratio can improve the initial setting time of the MPC paste while exceeding the 5% can reduce the initial setting time. Also, the mechanical

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behavior of the hardened MPC pastes is affected by the boric acid content. Boric acid can reduce the compressive strength when added to an MPC paste with a low M/P ratio. Whereas, the reduction of the compressive strength with the addition of boric acid becomes marginal for MPC pastes with high M/P molar ratio.

MPC pastes were found to be very susceptible to water content, unlike OPC pastes. Adding or reducing a small amount of water can change its fluidity considerably. For the M/P ratio of 3, increasing the W/P ratio by 18% can increase the fluidity by more than 70%.

It was also found that the optimal M/P molar ratio is affected by the W/P ratio and vice versa. That is, the experimental results demonstrated that for each W/P molar ratio, there is an optimal M/P molar ratio; and for each M/P molar ratio, there is an ideal W/P molar ratio. The boric acid content can change the impact of the water content on the mechanical behavior of the MPC paste. For example, with M/P ratio of three, an increase in W/P molar ratio by 18% reduces the compressive strength by 19%. Adding 5% of boric acid to the same mixtures can reduce the compressive strength by only 4%.

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